

AWS Cloud – Fresher Interview

Questions & Answers

Topics: IAM | EBS | EFS | S3 | Organizations | VPC | Load Balancer | Auto Scaling | RDS | SQS

■ IAM

Q1. Can a user with IAM access delete other users or even an admin? Explain.

Yes, it depends entirely on the permissions granted to that IAM user. If a user has been granted the `iam:DeleteUser/Administrator` access permission, they can delete other users, including admin accounts. However, AWS follows the Principle of Least Privilege, meaning users should only have the minimum permissions required. In practice, only root accounts or highly privileged admin roles should have `iam:DeleteUser` rights.

Q2. What is the difference between an IAM Role and an IAM User?

An IAM User is a permanent identity with long-term credentials (username/password or access keys) used by a specific person or application. An IAM Role is a temporary identity that can be assumed by AWS services, users, or external identities (via federation). Roles do not have static credentials; they issue temporary security tokens via AWS STS (Security Token Service). Best practice recommends using roles over users wherever possible — especially for EC2 instances, Lambda functions, and cross-account access.

Q3. What is an IAM Policy? What are the types of IAM policies?

An IAM Policy is a JSON document that defines permissions — what actions are allowed or denied on which AWS resources. Types include: (1) Identity-based policies — attached to users, groups, or roles; (2)

Resource-based policies — attached directly to resources like S3 buckets or SQS queues; (3) Permission Boundaries — set the maximum permissions an identity-based policy can grant; (4) Service Control Policies (SCPs) — used in AWS Organizations to restrict permissions across accounts; (5) Session policies — passed when assuming a role to limit its permissions.

■ EBS & EFS

Q4. Does EBS have storage classes? What is the minimum EBS size requirement?

No, EBS does not have 'storage classes' like S3. Instead, EBS offers different volume types optimized for different workloads: (1) gp2/gp3 – General Purpose SSD; (2) io1/io2 – Provisioned IOPS SSD for high-performance databases; (3) st1 – Throughput Optimized HDD for big data workloads; (4) sc1 – Cold HDD for infrequently accessed data. The minimum EBS volume size is 1 GiB for SSD volumes (gp2, gp3, io1, io2) and 125 GiB for HDD volumes (st1, sc1).

Q5. Compare EBS and EFS across five key points.

(1) Attachment: EBS can be attached to a single EC2 instance at a time (except Multi-Attach for io1/io2); EFS can be mounted simultaneously by thousands of EC2 instances. (2) File System: EBS is a block storage device; EFS is a fully managed NFS (Network File System). (3) Scalability: EBS size must be manually provisioned and resized; EFS scales automatically with no pre-provisioning needed. (4) Use Case: EBS is ideal for databases, boot volumes, and single-instance workloads; EFS suits shared file storage, CMS platforms, and containers. (5) Pricing: EBS charges per provisioned GB; EFS charges per GB stored.

■ S3 – Simple Storage Service

Q6. What are the storage classes available in Amazon S3?

Amazon S3 provides the following storage classes: (1) S3 Standard – for frequently accessed data with high availability; (2) S3 Intelligent-Tiering – automatically moves data between tiers based on access patterns; (3) S3 Standard-IA (Infrequent Access) – for data accessed less frequently but requiring rapid retrieval; (4) S3 One Zone-IA – lower-cost IA storage in a single AZ; (5) S3 Glacier Instant Retrieval – archival data with millisecond access; (6) S3 Glacier Flexible Retrieval – archival with retrieval times of minutes to hours; (7) S3 Glacier Deep Archive – lowest-cost storage for long-term retention.

Q7. What are the key advantages of Amazon S3?

Key advantages of Amazon S3 include: (1) Durability – 99.999999999% (11 nines) durability by redundantly storing data across multiple AZs; (2) Scalability – virtually unlimited storage with no pre-provisioning; (3) Security – supports encryption at rest (SSE-S3, SSE-KMS, SSE-C) and in transit (TLS/HTTPS); (4) Cost-effectiveness – multiple storage tiers to match access patterns and costs; (5) Accessibility – objects accessible via HTTPS URLs, SDKs, CLI, or the console globally; (6) Lifecycle Policies – automate data movement between storage classes or deletion; (7) Static website hosting – S3 can serve static websites directly.

Q8. What are the types of replication available in S3?

S3 offers two replication types: (1) Cross-Region Replication (CRR) – automatically replicates objects from a source bucket in one AWS Region to a destination bucket in a different Region. Used for compliance, latency reduction, and disaster recovery. (2) Same-Region Replication (SRR) – replicates objects between buckets within the same AWS Region. Used for log aggregation, live replication between production and test environments,

and data sovereignty compliance. Both require versioning to be enabled on source and destination buckets.

Q9. Why is versioning important in S3? Explain its significance.

Versioning in S3 allows you to preserve, retrieve, and restore every version of every object in a bucket. Its importance lies in: (1) Accidental deletion protection – a delete operation adds a delete marker rather than permanently removing the object; (2) Overwrites protection – all previous versions are retained even if an object is overwritten; (3) Compliance – helps meet regulatory requirements for data retention; (4) Disaster recovery – easy rollback to any previous version of an object; (5) Audit trail – maintain a full history of changes to objects over time.

Q10. Write a sample S3 bucket policy JSON to restrict access to specific users only.

Need to perform your self

Q11. What are Pre-signed URLs in S3? Why are they used and what is their importance?

A Pre-signed URL is a time-limited URL that grants temporary access to a private S3 object without requiring the requester to have AWS credentials. The URL is generated by an authorized AWS user and embeds authentication information, an expiration timestamp, and the permitted action (GET or PUT). Why we use them: (1) Share private files securely with external users or applications without making the bucket public; (2) Allow users to upload files directly to S3 without exposing AWS credentials; (3) Limit access to a specific time window. Importance: They enable secure, temporary, and fine-grained object-level access, making them critical for file-sharing applications, media streaming, and secure document distribution.

■ AWS Organizations & Landing Zone

Q12. What challenges do customers face as they scale their AWS usage, and how does AWS Organizations help?

Challenges include: (1) Account sprawl – managing hundreds of accounts with no central oversight; (2) Security inconsistencies – different teams applying different security configurations; (3) Cost visibility – difficulty tracking spend across teams and projects; (4) Compliance – enforcing organizational policies uniformly. AWS Organizations addresses these by providing a management hierarchy (root, OUs, accounts), centralized billing with consolidated billing, Service Control Policies (SCPs) to enforce permission guardrails, and integration with AWS Control Tower for governance at scale.

Q13. What does AWS Organizations manage, and what are its best practices?

AWS Organizations manages: (1) Account hierarchy through Organizational Units (OUs); (2) Centralized billing across all member accounts; (3) SCPs to control maximum permissions; (4) Integration with AWS SSO for unified sign-on; (5) Delegated administrator accounts for services like GuardDuty, Security Hub, etc. Best practices: (1) Use a dedicated management (payer) account with no workloads; (2) Group accounts into OUs by function (Dev, Staging, Prod, Security, Logging); (3) Enforce SCPs at OU level to restrict risky actions (e.g., deny disabling CloudTrail); (4) Enable AWS CloudTrail organization-wide; (5) Use tag policies for cost allocation.

Q14. What is an AWS Landing Zone?

An AWS Landing Zone is a pre-configured, secure, multi-account AWS environment based on AWS best practices. It establishes a baseline architecture that includes: (1) A management account for billing and governance; (2) A Log Archive account for centralized logging (CloudTrail,

Config); (3) An Audit/Security account for security tooling; (4) A shared services account for common resources (DNS, directory).

AWS Control Tower is the managed service used to set up and govern a Landing Zone. It provides guardrails (mandatory and optional) as SCPs and AWS Config rules, and uses Account Factory to vend new accounts consistently.

■ VPC, Subnets, Security & Networking

Q15. What is CIDR? What are VPC Endpoints, and why are they important?

CIDR stands for Classless Inter-Domain Routing. It is a method for allocating IP addresses using a notation like 10.0.0.0/16, where /16 indicates that the first 16 bits are the network prefix, allowing 65,536 IP addresses. VPC Endpoints are private connections between your VPC and supported AWS services (S3, DynamoDB, etc.) that do not require internet gateways, NAT devices, or VPN connections. Types: (1) Gateway Endpoints – for S3 and DynamoDB (free); (2) Interface Endpoints (powered by AWS PrivateLink) – for most other services, using an ENI with a private IP. Importance: They improve security by keeping traffic within the AWS network and reduce data transfer costs.

Q16. What do you understand by Subnets in AWS VPC?

A Subnet is a range of IP addresses within a VPC. Subnets allow you to partition the VPC network for different types of resources. Types: (1) Public Subnet – has a route to an Internet Gateway; instances can have public IPs and receive inbound traffic from the internet; (2) Private Subnet – no route to the Internet Gateway; instances communicate with the internet via a NAT Gateway for outbound traffic only; (3) Isolated/DB Subnet – used for databases with no internet access at all. Subnets are Availability Zone-specific, meaning each subnet resides in exactly one AZ, enabling high-

availability architectures by distributing resources across multiple subnets in different AZs.

Q17. Explain the difference between NACLs and Security Groups.

Security Groups operate at the instance (ENI) level, are stateful (return traffic is automatically allowed), support Allow rules only, and are evaluated as a collective set for all rules. NACLs (Network Access Control Lists) operate at the subnet level, are stateless (return traffic must be explicitly allowed), support both Allow and Deny rules, and rules are evaluated in numerical order (lowest first). Key use case difference: Security Groups are your first line of defense for instance-level traffic control; NACLs provide an additional layer for subnet-wide traffic filtering, useful for blocking specific IP ranges at scale.

Q18. What is a Bastion Host? If VPC A1 is connected to B1 and A1 to C1, are B1 and C1 connected?

A Bastion Host (also called a Jump Server) is a hardened EC2 instance placed in a public subnet that provides secure SSH/RDP access to instances in private subnets. Administrators connect to the Bastion Host over the internet, then hop to private instances. It reduces the attack surface by centralizing access control and logging. Regarding VPC peering transitivity: No, B1 and C1 are NOT connected. VPC Peering is non-transitive. Even though A1 is peered with both B1 and C1, traffic cannot flow from B1 to C1 through A1. A direct peering connection between B1 and C1 would need to be established separately. For transitive routing, AWS Transit Gateway should be used instead.

■ Load Balancers, Auto Scaling & Service Quotas

Q19. Name some open-source load balancers and list the types of load balancers in AWS.

Popular open-source load balancers include: HAProxy, NGINX, Traefik, Envoy, and Caddy. AWS provides three types of managed load balancers under Elastic Load Balancing (ELB): (1) Application Load Balancer (ALB) – Layer 7 (HTTP/HTTPS), supports path-based and host-based routing, WebSockets, and Lambda targets; (2) Network Load Balancer (NLB) – Layer 4 (TCP/UDP/TLS), ultra-high performance, static IP support, ideal for real-time gaming and financial applications; (3) Gateway Load Balancer (GWLB) – Layer 3/4, used to deploy, scale, and manage third-party virtual network appliances like firewalls and intrusion detection systems.

Q20. What are Auto Scaling Group policies? Explain the types.

Auto Scaling Group (ASG) policies define how the group scales in response to demand. Types include: (1) Target Tracking Scaling – the most recommended; automatically adjusts capacity to maintain a target metric value (e.g., keep average CPU at 50%); (2) Step Scaling – scales based on the magnitude of a CloudWatch alarm breach, adding/removing instances in steps; (3) Simple Scaling – triggers a single scaling action after a cooldown period when an alarm breaches a threshold; (4) Scheduled Scaling – scales at specific times based on predictable load patterns (e.g., scale up every Monday 9 AM); (5) Predictive Scaling – uses ML to forecast future traffic and proactively adjusts capacity.

Q21. What is a Service Quota in AWS? Why does it matter?

A Service Quota (formerly known as Service Limit) is the maximum value for a resource, action, or item in your AWS account for a given Region. Examples: maximum number of EC2 instances per region, maximum VPCs per account, maximum Lambda concurrent executions. They matter because: (1) They prevent accidental over-provisioning and runaway costs; (2) They ensure fair resource distribution across all AWS customers; (3) Understanding them is critical for production planning — breaching a quota causes service failures. Quotas can be viewed and increase requests

submitted via the AWS Service Quotas console. Some quotas are soft (adjustable) and others are hard (fixed).

Q22. Why do we use Read Replicas in RDS? Name three alternative services to Amazon SQS.

Read Replicas in Amazon RDS are used to: (1) Offload read-heavy workloads from the primary (write) database instance, improving overall throughput; (2) Serve reporting and analytics queries without impacting the production database; (3) Support geographic distribution – replicas can be created in different Regions to reduce read latency for global users; (4) Act as a standby that can be promoted to a primary in a failover scenario (though Multi-AZ is preferred for automatic failover). Three alternatives to Amazon SQS: (1) Apache Kafka – distributed event streaming platform for high-throughput messaging; (2) RabbitMQ – open-source message broker supporting multiple protocols (AMQP, STOMP, MQTT); (3) ActiveMQ – Apache's open-source message broker, also available as Amazon MQ (managed service).

*End of AWS Fresher Interview Q&A — Best
of luck with your AWS journey! ■■*